

TARKO-SALE, Russian Federation

WMO: 235520

Lat: 64.9114N

Lon: 77.7708E

Elev: 27

StdP: 101.00

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.8	(c) -41.1	(d) -45.7	(e) 0.0	(f) -42.9	(g) -43.3	(h) 0.1	(i) -40.7	(j) 7.7	(k) -14.1	(l) 6.7	(m) -12.9	(n) 1.3	(o) 70	(p) 0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	28.2	19.1	25.8	17.8	23.5	16.8	19.9	25.9	18.7	24.2	17.5	22.4	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	22.4	16.6	11.8	21.1	15.3	10.9	20.3	57.2	26.0	53.3	24.2	49.5	22.4	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.2	5.3	-44.9	30.4	4.2	1.9	-47.9	31.8	-50.4	32.9	-52.8	34.0	-55.9	35.4		
			-44.1	21.7	3.4	1.8	-46.5	23.0	-48.5	24.1	-50.4	25.1	-52.9	26.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.9	-24.7	-22.2	-13.2	-7.0	1.0	12.5	16.3	12.5	5.9	-3.4	-16.7	-21.2	(6)
(7)		DBStd	15.93	9.81	9.29	8.12	7.44	6.11	6.10	4.56	3.92	4.48	6.37	9.67	10.11	(7)
(8)		HDD10.0	5879	1077	903	718	511	287	44	3	16	138	414	800	967	(8)
(9)		HDD18.3	8542	1335	1136	977	761	538	193	96	187	373	673	1050	1225	(9)
(10)		CDD10.0	434	0	0	0	0	8	118	199	93	16	0	0	0	(10)
(11)		CDD18.3	56	0	0	0	0	1	17	33	6	0	0	0	0	(11)
(12)		CDH23.3	518	0	0	0	0	3	166	301	47	1	0	0	0	(12)
(13)		CDH26.7	128	0	0	0	0	1	36	84	7	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.2	2.3	2.6	2.9	3.0	2.7	2.5	2.3	2.4	2.6	2.4	2.3	(14)
(15)	Precipitation	PrecAvg	514	27	22	27	31	37	52	61	75	59	53	39	31	(15)
(16)		PrecMax	762	50	50	61	63	86	114	166	149	116	124	71	60	(16)
(17)		PrecMin	370	8	3	5	6	15	6	3	13	17	21	13	14	(17)
(18)		PrecStd	82	10	10	12	14	17	26	34	35	25	19	14	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.6	-1.5	2.7	9.9	22.2	30.0	31.4	27.4	20.9	11.1	2.2	-0.3	(19)
(20)			MCWB	-3.1	-2.1	0.5	5.4	14.1	19.0	20.0	20.7	15.1	8.8	1.5	-0.8	(20)
(21)		2%	DB	-5.5	-4.2	0.7	6.7	17.3	27.6	29.1	24.4	17.6	8.0	0.9	-2.7	(21)
(22)			MCWB	-6.0	-4.8	-0.7	3.4	11.5	18.5	19.1	18.1	13.4	6.7	0.4	-3.2	(22)
(23)		5%	DB	-8.8	-6.5	-0.5	4.7	13.5	25.3	26.7	21.5	15.0	6.0	-0.7	-4.4	(23)
(24)			MCWB	-9.2	-7.1	-1.6	2.1	9.1	17.3	18.2	16.5	12.1	4.9	-1.2	-4.8	(24)
(25)		10%	DB	-11.2	-9.0	-2.1	3.2	10.2	22.7	24.4	19.1	12.7	4.3	-2.8	-7.0	(25)
(26)			MCWB	-11.6	-9.5	-3.1	1.3	6.9	16.0	17.4	15.4	10.4	3.4	-3.3	-7.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.1	-1.9	0.8	5.6	14.5	20.7	21.8	21.5	16.4	9.2	1.7	-0.7	(27)
(28)			MCDWB	-2.8	-1.6	2.0	8.8	20.1	27.3	28.4	26.3	20.1	10.8	2.1	-0.3	(28)
(29)		2%	WB	-6.0	-4.6	-0.4	3.6	12.0	19.4	20.2	18.8	14.0	6.8	0.5	-3.1	(29)
(30)			MCDWB	-5.6	-4.1	0.6	6.1	16.6	25.6	26.7	22.6	16.8	7.8	0.9	-2.7	(30)
(31)		5%	WB	-9.1	-7.0	-1.6	2.5	9.5	18.1	19.1	17.2	12.4	5.1	-1.2	-4.9	(31)
(32)			MCDWB	-8.8	-6.4	-0.7	4.4	13.3	24.2	24.9	20.9	14.9	6.0	-0.7	-4.4	(32)
(33)		10%	WB	-11.5	-9.5	-3.1	1.4	6.9	16.6	18.0	15.7	10.5	3.4	-3.3	-7.3	(33)
(34)			MCDWB	-11.2	-9.0	-2.1	3.0	10.1	21.8	23.2	18.8	12.5	4.3	-2.7	-6.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	7.9	9.4	9.6	7.7	8.7	8.8	7.6	6.1	5.1	7.0	7.4	(35)
(36)			MCDBR	9.0	8.4	8.9	7.9	12.4	12.2	12.2	11.6	10.0	5.7	7.2	7.2	(36)
(37)			MCWBR	9.0	8.1	7.9	5.6	7.6	5.3	5.0	6.0	6.0	4.6	7.2	7.2	(37)
(38)		5% WB	MCDBR	9.1	8.5	8.7	7.8	12.1	11.8	10.7	10.4	9.4	5.5	7.1	7.4	(38)
(39)			MCWBR	9.1	8.3	8.0	6.0	7.8	6.0	5.0	5.8	6.0	4.6	7.2	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.213	0.272	0.324	0.342	0.358	0.375	0.418	0.397	0.338	0.292	0.208	0.219	(40)	
(41)		taud	1.876	2.026	2.001	2.058	2.221	2.398	2.260	2.290	2.418	2.196	1.905	1.881	(41)	
(42)		Ebn at Noon	271	592	721	813	842	839	780	758	737	573	267	60	(42)	
(43)		Edh at Noon	26	64	105	126	119	103	115	101	72	54	23	9	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.14	0.84	2.48	4.58	5.44	5.42	4.95	3.35	1.95	0.69	0.23	0.05	(44)	
(45)		RadStd	0.01	0.07	0.16	0.43	0.42	0.41	0.56	0.41	0.28	0.08	0.04	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+1.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page